<table>
<tr><td>PSNo.</td><td>26</td></tr>
<tr><td>ProjectTitle</td><td>ParkingManagementSystem</td></tr>
<tr><td>Problem Statement</td><td>Buildaconsoleparkingsystemforasmalllotwherevehiclesareparkedinnumberedslots,removedonexit,and charged according to the vehicle type and the parking time.
Language:Python3Concepts:Lists,Dictionaries,Conditions,File Handling
Note:Thelothas5slots.Rates:carRs.20perhour,bikeRs.10perhour.Hoursareroundedup andtheminimum chargeis1 hour.Dataissaved inparking.txt. Menu:1Entry,2 Exit,3Status.Each testcasestartswith an empty lot unless stated.</td></tr>
<tr><td>Test Cases / CaseStatements (4 cases)</td><td>
<table>
<tr><th>INPUT</th><th>EXPECTEDOUTPUT</th></tr>
<tr><td colspan="2">TestCase1:NormalInput</td></tr>
<tr><td>Menu option 1 (Entry)
Vehicle:MH12AB1234
Type: car</td><td>VehicleMH12AB1234parkedinslot1.
Slots free: 4</td></tr>
<tr><td colspan="2">TestCase2:Boundary/MinimumInput</td></tr>
<tr><td>MH12AB1234(car)isinslot1
Menu option 2 (Exit)
Vehicle:MH12AB1234
Hours: 0.5</td><td>Hourscharged:1(minimum) | Fee:20 Slot
1 is now free.</td></tr>
<tr><td colspan="2">TestCase3:InvalidInput</td></tr>
<tr><td>MH12AB1234isparked
(a) Entry:MH12AB1234
(b) Entry:KA01XY9999,Type: truck
(c) Exit: XX00XX0000</td><td>(a) VehicleMH12AB1234isalreadyparked.
(b) Invalidtype.Entercaror bike.
(c) VehicleXX00XX0000notfound.</td></tr>
<tr><td colspan="2">TestCase4:Different ValidScenario</td></tr>
<tr><td>(a) Entry:KA01AB1111,Type: bike
(b) Exit:KA01AB1111,Hours:2.5
(c) Option3(Status)</td><td>(a) VehicleKA01AB1111parkedinslot1.
(b) Hourscharged:3(roundedup) | Fee:30
(c) Slotsfree:5of5</td></tr>
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</td></tr>
<tr><td>BareMinimum Requirements</td><td>1. WritethesolutioninPython3(one.pyfile) and run itfromtheconsole.
2. Storetheslotsin alistandeach parked vehicleasa dictionary.
3. Writefunctionsforentry,exit,feecalculationandstatus.
4. Useconditionsforthevehicletype,thefull-lotcheckandtheminimumcharge rule.
5. Validatethevehiclenumber,thetypeand thehours,and savethedatato afileafterevery change.</td></tr>
<tr><td>Instruction</td><td>1. Understandtheproblemstatement anddefinetheproblemclearly.
2. Developthesolutionusingproperlogicandprogramming concepts.
3. Implementallrequiredfeaturesandpassthegiventestcases.
4. Testanddebugtheprojecttoensurecorrect output.
5. Preparethedocumentation/reportanduploadthecodeandreportto GitHub.
6. Completeandsubmittheprojectwithin thegiven deadline.</td></tr>
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